Totally equimodular matrices: decomposition and triangulation

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Université Libre de Bruxelles[1]

Joint work with: Patrick Chervet and Roland Grappe

Seminar in OR – Thursday 30 Oct.

Linear optimization problem

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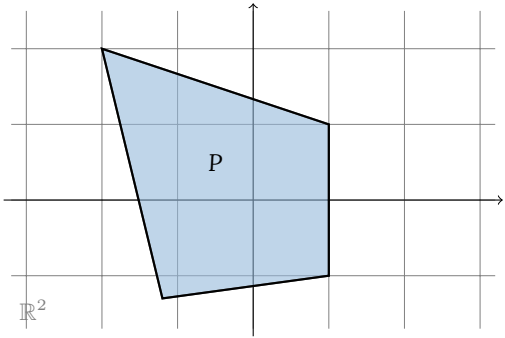

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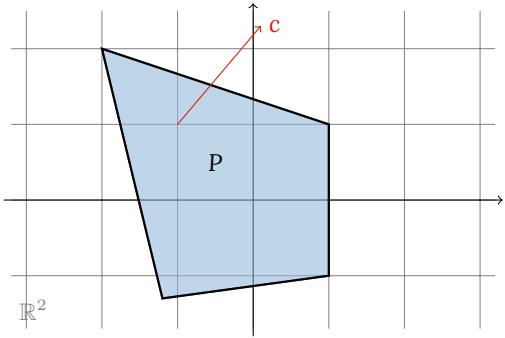

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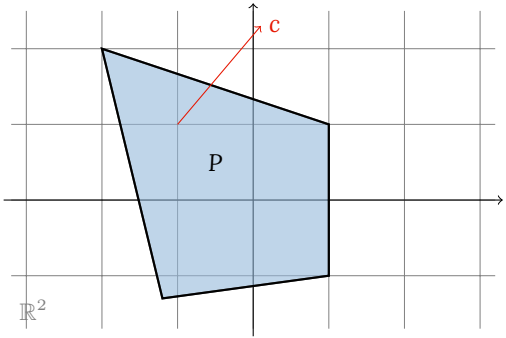

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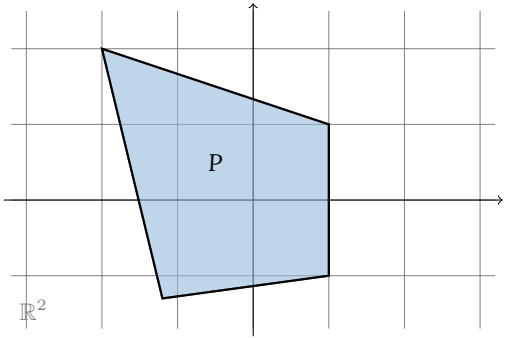

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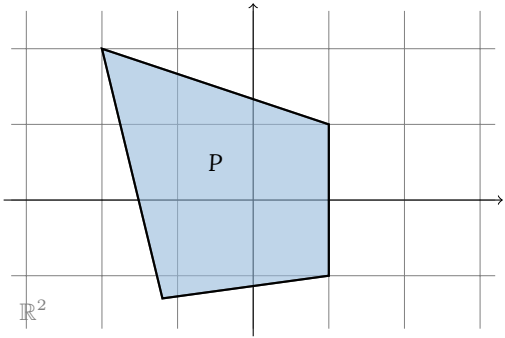

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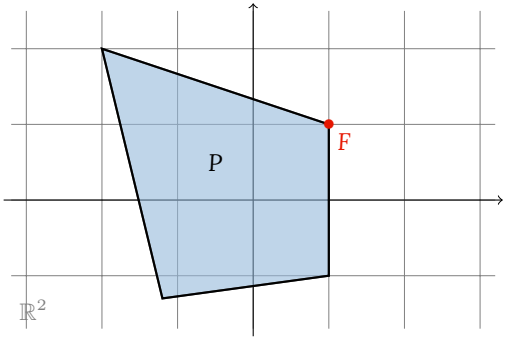

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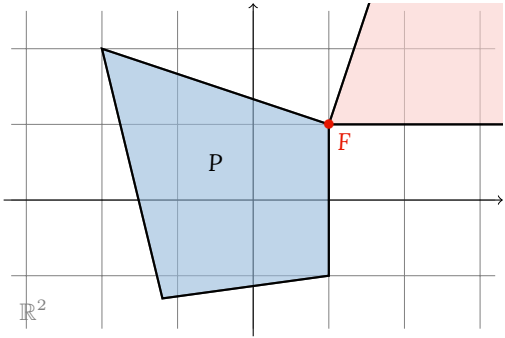

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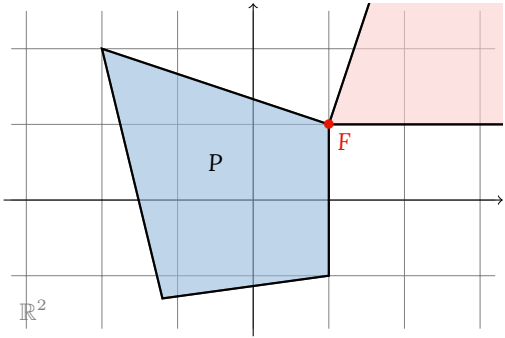

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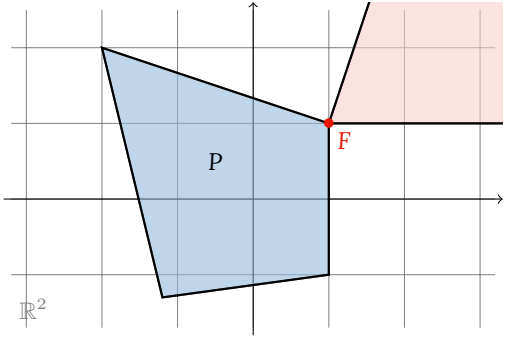

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Note: we can suppose c integer

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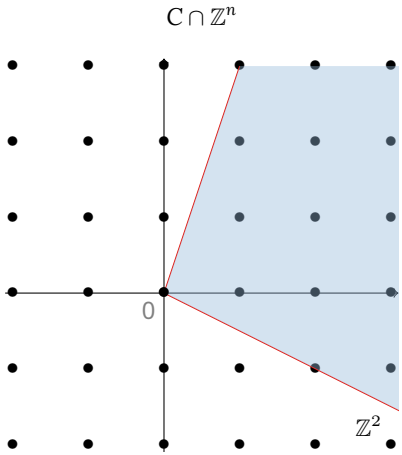

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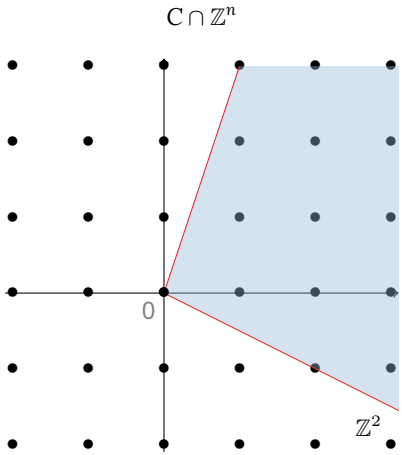

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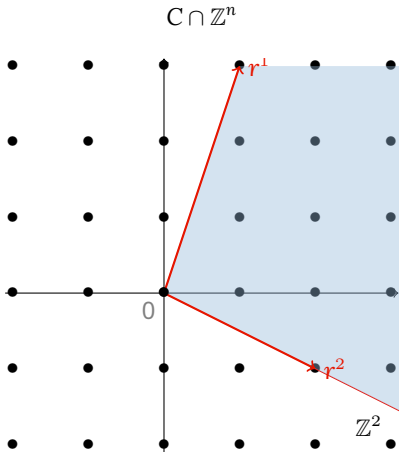

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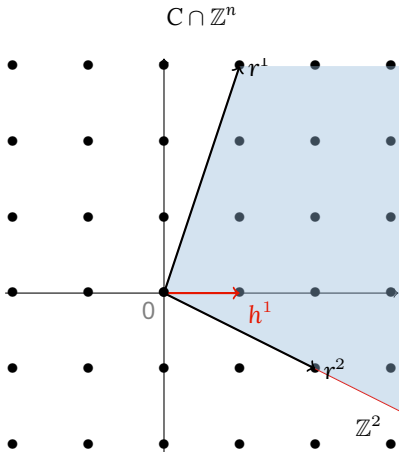

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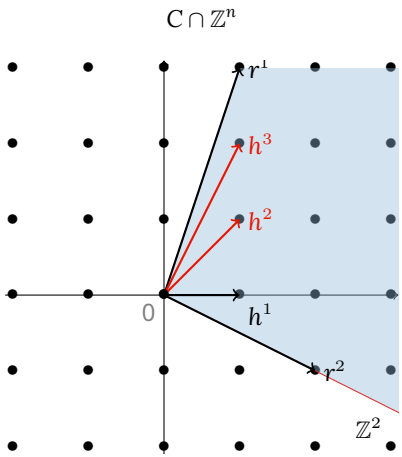

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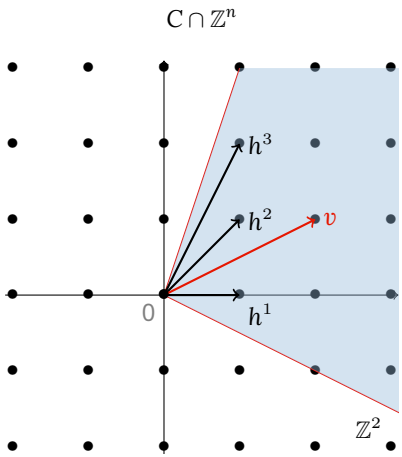

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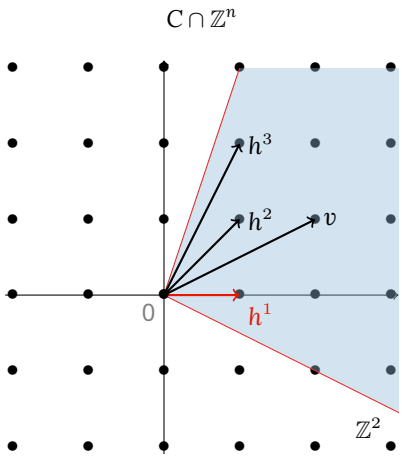

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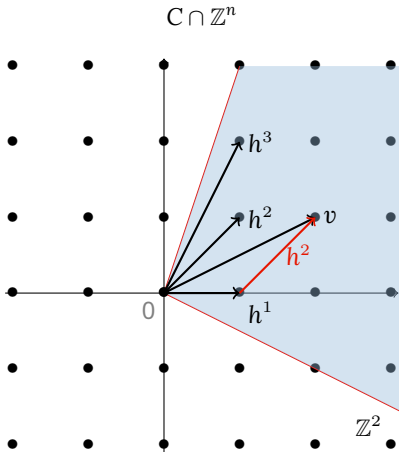

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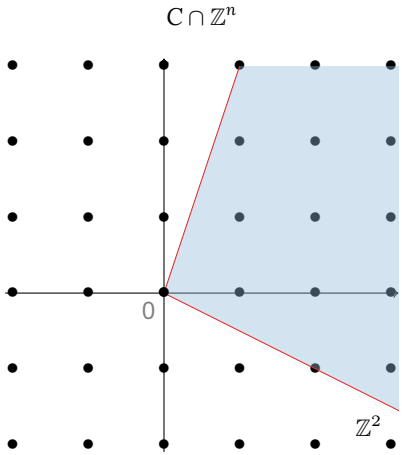

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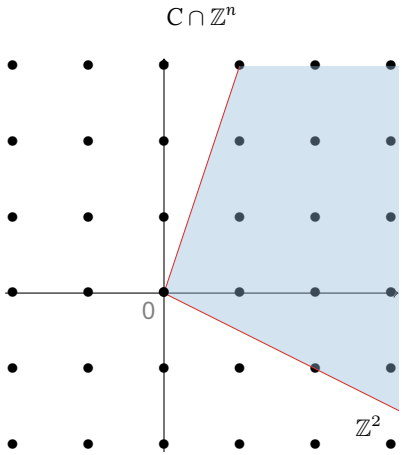

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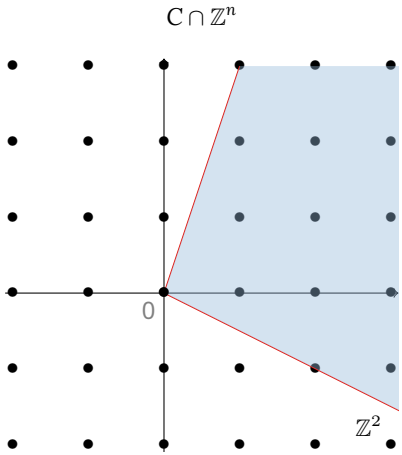

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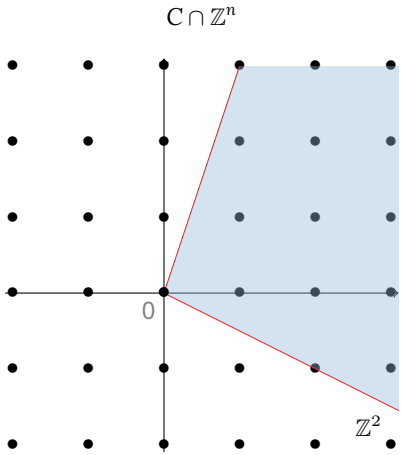

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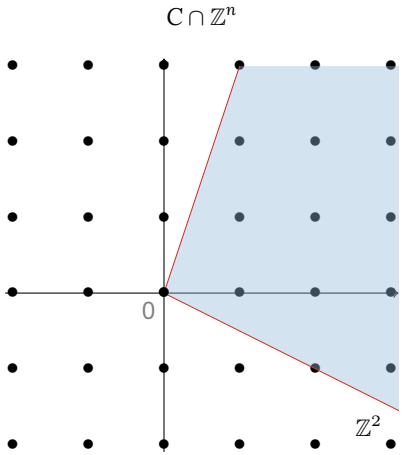

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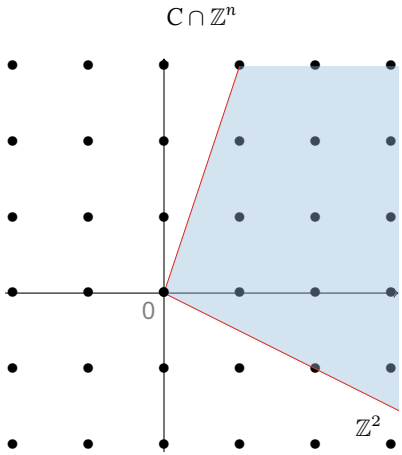

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If n suffice, then C satisfies the **Integer Carathéodory property**.

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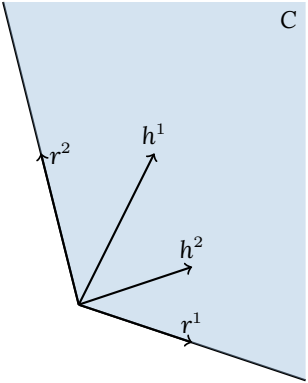

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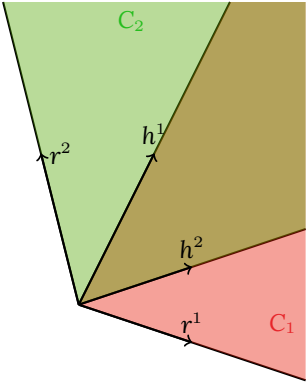

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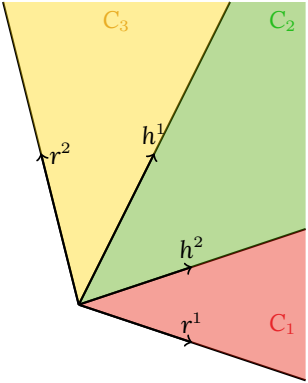

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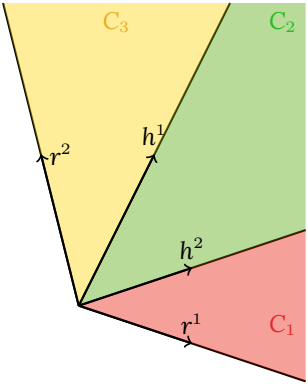

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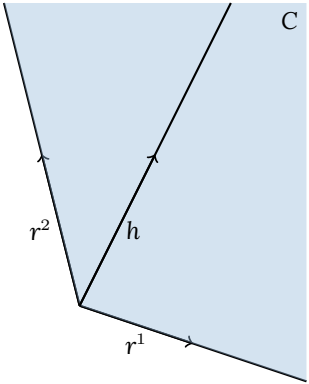

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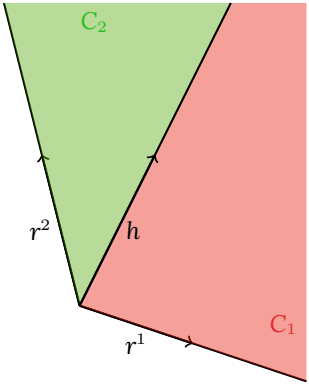

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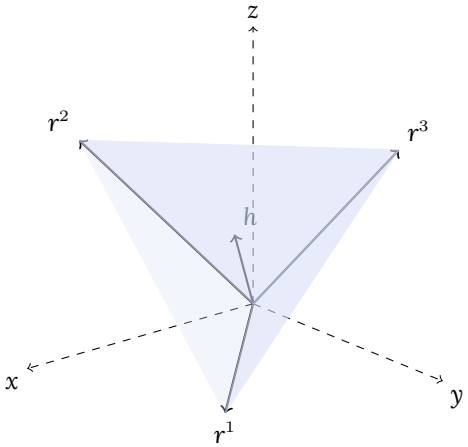

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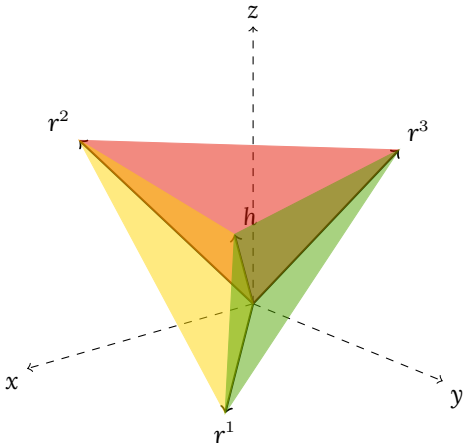

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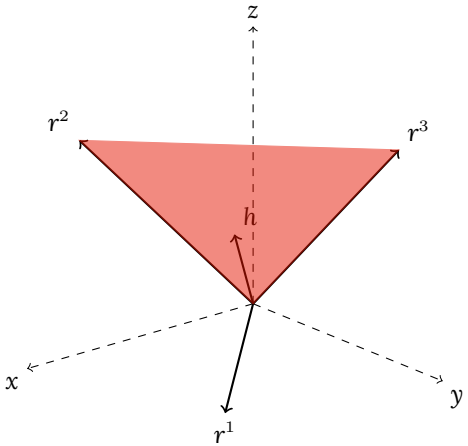

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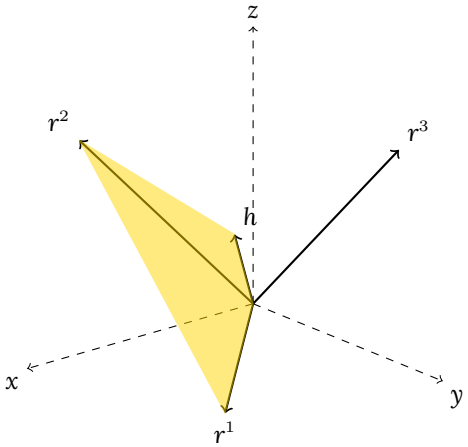

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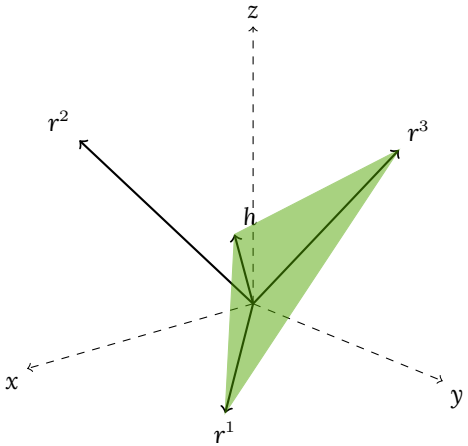

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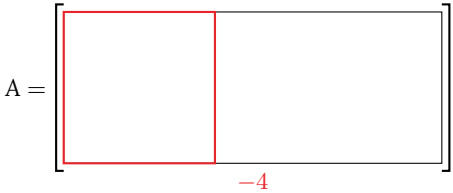

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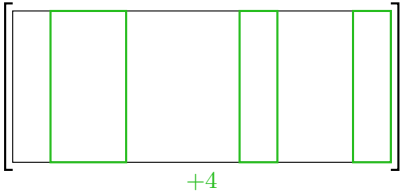

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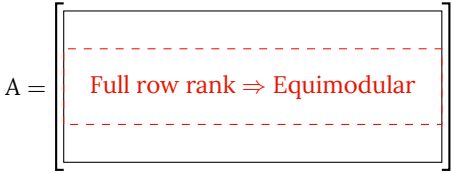

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tu-set totally unimodular				

Building blocks of linearly independent te-sets with r rows

Type of te-sets	det	examples	Hilbert basis of cone(A)	unim. Hilbert triang.
tu-set totally unimodular	1			

Building blocks of linearly independent te-sets with r rows

Type of te-sets	det	examples	Hilbert basis of cone(A)	unim. Hilbert triang.
tu-set totally unimodular	1	$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$		

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te-lace not a tu-set, every proper subset is a tu-set	2			

Building blocks of linearly independent te-sets with r rows

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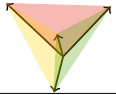
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$A \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 

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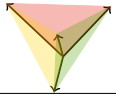
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$\mathrm{A} \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
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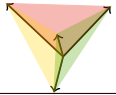
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$\mathrm{A} \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
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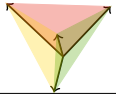
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$A \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
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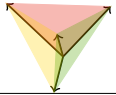
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$A \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
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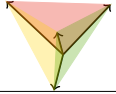
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$A \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
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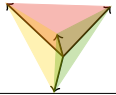
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$\mathrm{A} \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
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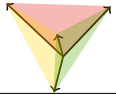
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$A \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
te-interlace not a tu-set, every pair forms a te-lace	**thin:** 2^{r-1}	$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & 0 & -1 \end{bmatrix}$	$A \cup \{\frac{1}{2}(a_i + a_j)\}_{i \neq j}$	triangulation of the second hypersimplex
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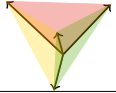
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$A \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
te-interlace not a tu-set, every pair forms a te-lace	**thin:** 2^{r-1}	$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & 0 & -1 \end{bmatrix}$	$A \cup \{\frac{1}{2}(a_i + a_j)\}_{i \neq j}$	triangulation of the second hypersimplex
	thick: 2^r	$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$	$A \cup \{\frac{1}{2}(a_i + a_j)\}_{i \neq j} \cup H$ for $H = \begin{cases} \{\frac{1}{4}\sum_i a_i\}, \text{ or} \\ \{\frac{3}{4}a_i + \frac{1}{4}\sum_{j \neq i} a_j\}_i. \end{cases}$	

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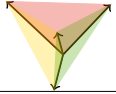
te-lace not a tu-set, every proper subset is a tu-set	2	$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$	$A \cup \{h = \frac{1}{2}\sum_i a_i\}$	stellar triangulation: $\{\mathsf{cone}(\{h\} \cup \{a_j\}_{j \neq i})\}_i$ 
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Main results and a conjecture

Theorem (Chervet, Grappe, V., 2025+)

Let A be a set of r linearly independent $0,\pm 1$ vectors. Then, A is a te-set if and only if it can be partitioned as follows.

$$A = \underbrace{U}_{\textit{tu-set}} \sqcup \underbrace{L_1 \sqcup \cdots \sqcup L_k}_{\textit{te-laces}} \sqcup \underbrace{S_1 \sqcup \cdots \sqcup S_\ell}_{\textit{thin te-interlaces}} \sqcup \underbrace{T_1 \sqcup \cdots \sqcup T_m}_{\textit{thick te-interlaces}}$$

where $U, L_1, \ldots, L_k, S_1, \ldots, S_\ell, \ldots, T_1, \ldots, T_m$ *are* ***mutually-tu****.*

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Let A be a linearly independent te-set with no thick te-interlaces of size 8 or more. Then $\mathsf{cone}(A)$ *has a* ***regular unimodular Hilbert triangulation****.*

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Theorem (Chervet, Grappe, V., 2025+)

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Conjecture (Chervet, Grappe, V., 2025+)

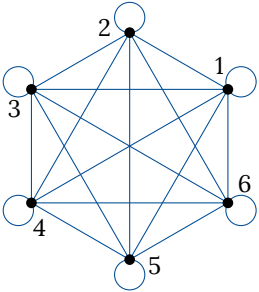
Up to resigning, there are only two thick te-interlaces:

$$(r = 4): \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix}, \text{ and}$$

$$(r = 6): \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 & -1 \end{bmatrix}.$$

Triangulation of thin te-interlaces

We embed $\mathring{K}_n$, the **complete graph** on n vertices with loops ii, as a convex n-gon in $\mathbb{R}^2$:



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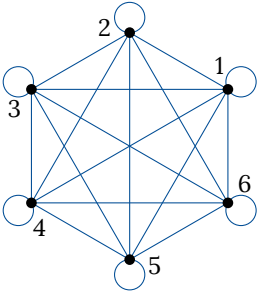

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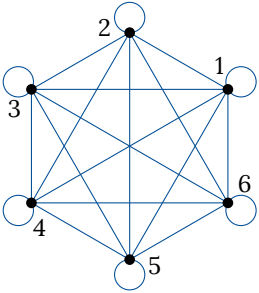

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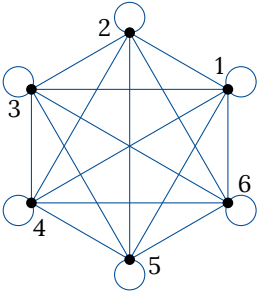

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Two distinct edges **intersect** if the associated segments or circles meet in $\mathbb{R}^2$.

Triangulation of thin te-interlaces

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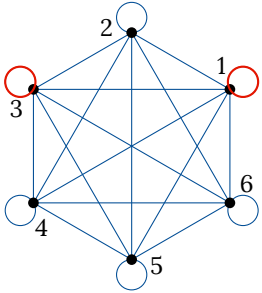

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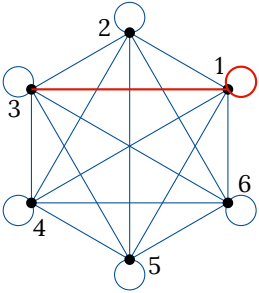

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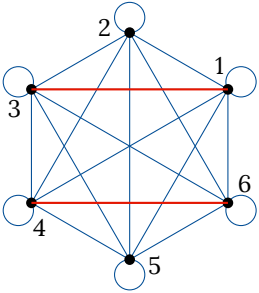

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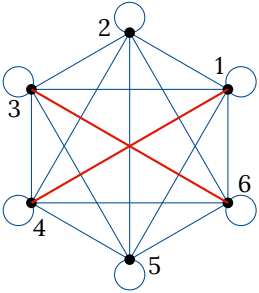

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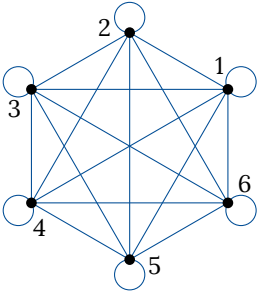

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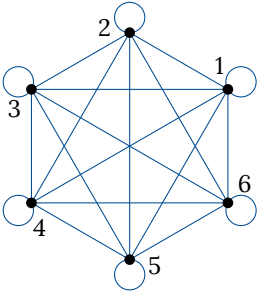

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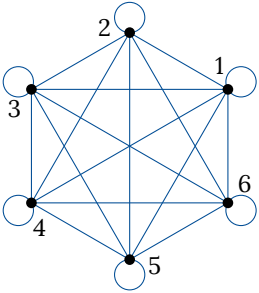

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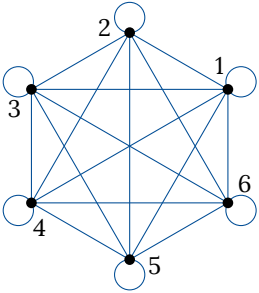

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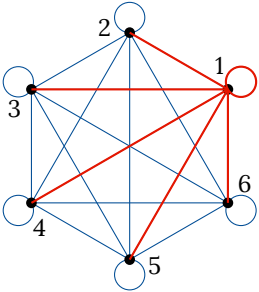

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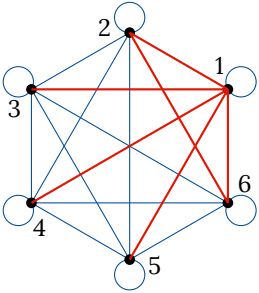

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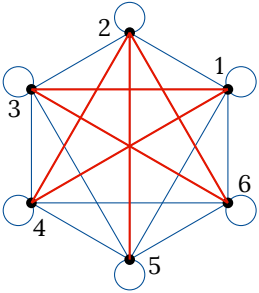

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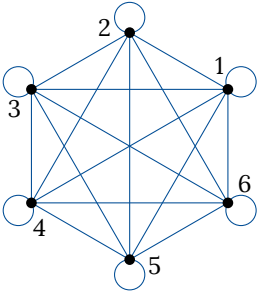

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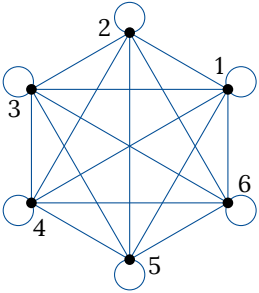

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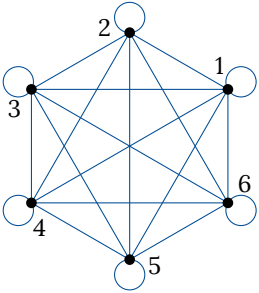

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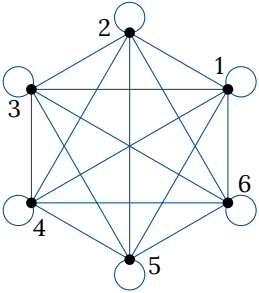

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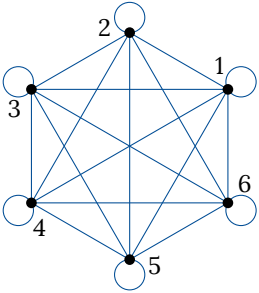

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Fact: The stellar cycles of $\mathring{K}_n$ describe a RUHT of $\mathsf{cone}(A)$, for A a thin te-interlace.

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Let A be a $0,1$ matrix and i a row index of A. The **row-i complement** $A_{[i]}$ of A is:

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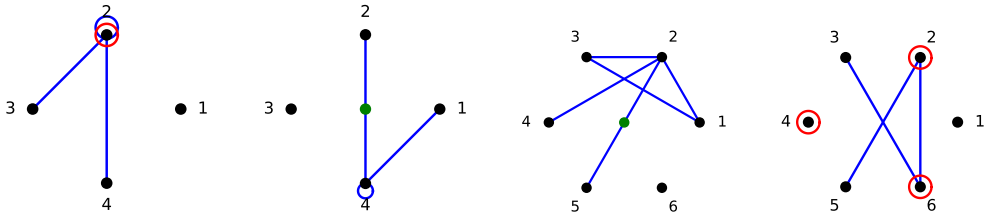

Thank you for listening!